# From Symmetric to Directed Membership: How the Attractive/Repulsive Forces Were Fixed

## 1 Where this comes from: UMAP is cross-entropy minimization

UMAP does not invent its attractive and repulsive forces by hand. It builds a *target* membership $\mu(i,j) \in [0,1]$ for every pair of points from the high-dimensional data (how strongly $j$ should be considered a neighbour of $i$), and a *model* membership

$$\nu(i,j) = \frac{1}{1 + a\,\rho(i,j)^{2b}}$$

from the current low-dimensional embedding, where $\rho(i,j)$ is the distance between $y_i$ and $y_j$. It then minimizes the cross-entropy between the two:

$$CE = \sum_{i,j} \underbrace{\mu(i,j)\log\frac{\mu(i,j)}{\nu(i,j)}}_{\text{attractive term}} \;+\; \underbrace{\big(1-\mu(i,j)\big)\log\frac{1-\mu(i,j)}{1-\nu(i,j)}}_{\text{repulsive term}}.$$

Taking the gradient of this loss with respect to the embedding positions is exactly where the attractive coefficient $\text{attr\_coeff}(\rho)$ and repulsive coefficient $\text{rep\_coeff}(\rho)$ come from. Concretely, each pair's contribution to the force is

$$\mu(i,j)\cdot\text{attr\_coeff}(\rho) \;-\; \big(1-\mu(i,j)\big)\cdot\text{rep\_coeff}(\rho).$$

So $\mu(i,j)$ is not a cosmetic weight – it is literally the coefficient the cross-entropy loss assigns to the attractive term for that pair. If $\mu(i,j)$ is high, $j$ pulls $i$ strongly; if it is low, $j$ mostly pushes $i$ away.

## 2 The problem: our data is directional, but $\mu$ was forced symmetric

In vanilla UMAP, each point $i$ first computes its own, one-sided view of the membership from its own $k$ nearest neighbours:

$$A(i,j) = \exp\!\left(-\frac{\max(0,\,D(i,j)-\rho_i)}{\sigma_i}\right),$$

using only row $i$ of the distance matrix $D$. This $A$ is not symmetric in general: $A(i,j) \neq A(j,i)$, because $i$ and $j$ may disagree on how close they are to each other. Vanilla UMAP then forces a single, shared value per edge by symmetrizing it with the probabilistic t-conorm,

$$\mu_{\text{sym}} = A + A^\top - A\cdot A^\top,$$

and uses $\mu_{\text{sym}}$ everywhere afterwards: in the cross-entropy weighting above, and in building the initial layout (the eigendecomposition used for spectral initialization requires a symmetric matrix).

This project's distance matrix $D_{\text{asym}}$ is asymmetric on purpose – it encodes a Randers (directional) field. The reconstructed distance in the embedding, $\rho_r(i \to j) = \|y_i - y_j\| + b_i \cdot (y_j - y_i)$, is *also* asymmetric on purpose, by the same Randers substitution. So we had an asymmetric target distance and an asymmetric reconstructed distance, but we were still comparing them through a forcibly symmetrized membership weight $\mu_{\text{sym}}$. That is the inconsistency: the one piece of the pipeline that was not allowed to be directional was exactly the piece deciding how strongly each direction should be pulled or pushed.

## 3 The fix, in two steps

**Step 1 – stop symmetrizing the force weight.** $A(i,j)$, the raw, one-sided membership described above, was already being computed. The only change needed was to use $A$ itself in the force computation instead of $\mu_{\text{sym}}$:

$$\text{force}(i,j) = A(i,j) \cdot \text{attr\_coeff}(\rho_r) \; - \; \big(1 - A(i,j)\big) \cdot \text{rep\_coeff}(\rho_r).$$

Now the cross-entropy sum runs over *ordered* pairs $(i,j)$, each with its own directed weight, instead of over undirected edges sharing one value. This is directionally consistent with both $D_{\text{asym}}$ and $\rho_r$. At this point, $\mu_{\text{sym}}$ was kept around for exactly one remaining job: feeding the initial layout, since that step's eigendecomposition still needed a symmetric input.

**Step 2 – remove the last symmetric matrix, from initialization too.** The initial layout does not have to come from a UMAP-style spectral (Laplacian-eigenmap) embedding at all. Real IsUMap – the method whose directional distance matrix this project builds on – initializes with classical (Torgerson) multidimensional scaling instead, applied directly to the distance matrix, with no membership graph and no symmetry requirement involved. Switching the initialization to classical MDS removed the only remaining reason $\mu_{\text{sym}}$ existed. It was deleted, along with the spectral initialization code that consumed it.

## 4 Result

There is now exactly one membership matrix in the pipeline, $A$, and it is never symmetrized. It is used only where the cross-entropy forces need it; initialization is built directly from the distance matrix and does not touch $A$ at all. The asymmetric target distance, the asymmetric reconstructed distance, and the directed force weighting are now consistent with each other end to end.